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Jhang et al.

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(54) **HEAD-MOUNTED DISPLAY DEVICE**

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Primary Examiner — William R Alexander

(51) **Int. Cl.**

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G02C 7/08 (2006.01)
G06F 1/16 (2006.01)

(52) **U.S. Cl.**

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(58) **Field of Classification Search**

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G02B 27/0176; G02B 27/0172; G02C
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1/163

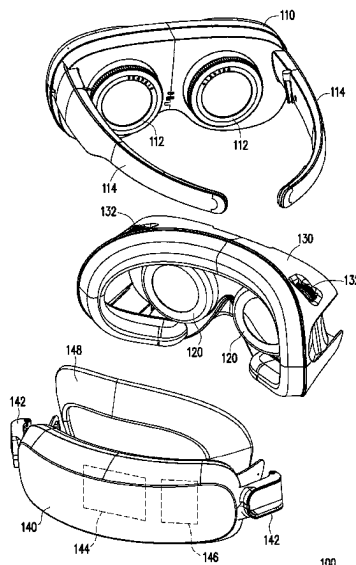
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See application file for complete search history.

(57) **ABSTRACT**

A head-mounted display device includes a body, two adjustment rings, and a face gasket. The body has two zoom lens groups corresponding to both eyes. The two adjustment rings are used to be detachably assembled to the corresponding zoom lens group, respectively. The face gasket is used to be detachably assembled to the body and used to contact a user's face. The face gasket has two adjustment knobs. Each of the adjustment knobs is used to couple the corresponding adjustment ring and rotate the corresponding zoom lens group for zooming, and a part of the each of the adjustment knobs is exposed to the outside for the user to manipulate.

6 Claims, 3 Drawing Sheets



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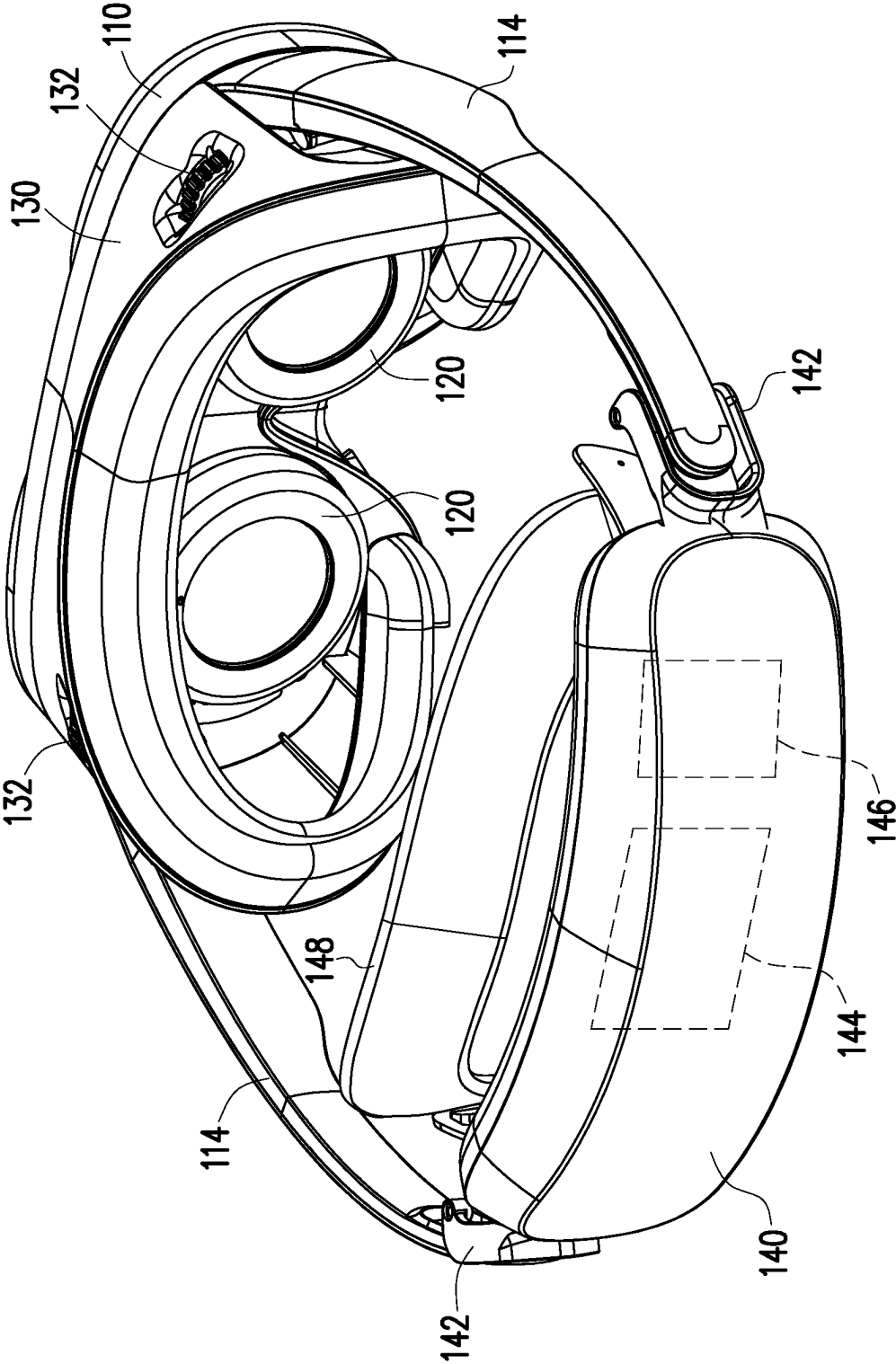


FIG. 1

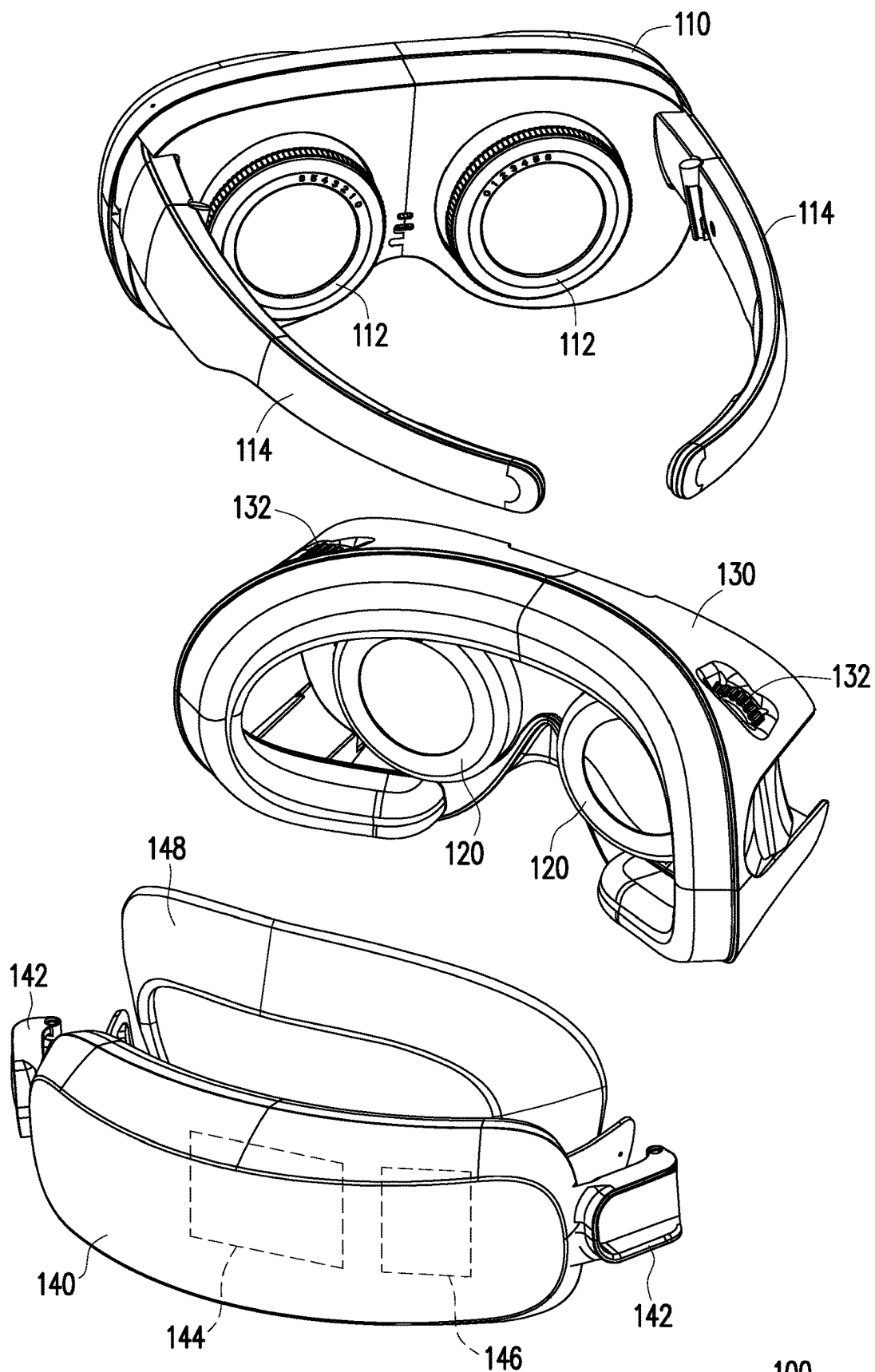


FIG. 2

100

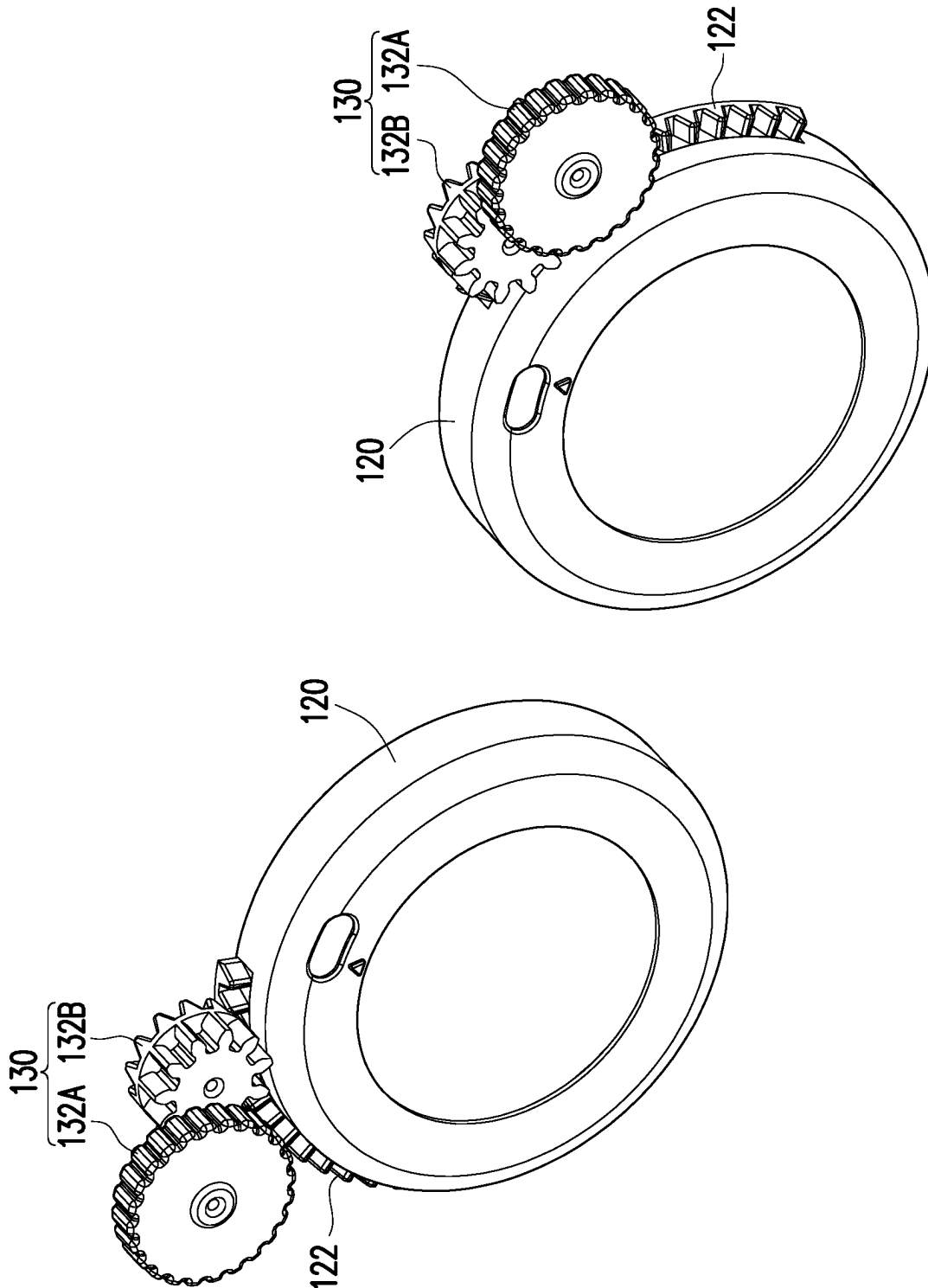


FIG. 3

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HEAD-MOUNTED DISPLAY DEVICE**CROSS-REFERENCE TO RELATED APPLICATION**

This application claims the priority benefit of U.S. provisional application Ser. No. 63/396,609, filed on Aug. 10, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND**Technical Field**

This application relates to a display device, and in particular to a head-mounted display device.

Description of Related Art

With the rapid progress of current technology, the types and functions of head-mounted display devices are becoming more and more diversified. For example, when a user wears this type of head-mounted display device, the gyroscope and position tracker inside the head-mounted display device will track the user's movement to project corresponding scene images, providing the user with a virtual world experience.

When using a head-mounted display device, since each user has a different degree of myopia, two zoom lens groups in the head-mounted display device are available to suit different users so that each user can get the best experience. Currently, head-mounted display devices mainly use manual zooming of the zoom lens groups. However, the installation of a face gasket for comfort and stability prevents the user from touching the zoom lens groups for zooming.

SUMMARY

This application provides a head-mounted display device, which may improve a problem that a face gasket obstructs zooming.

The head-mounted display device of this application includes a body, two adjustment rings, and a face gasket. The body has two zoom lens groups corresponding to both eyes. The two adjustment rings are used to be detachably assembled to a corresponding one of the zoom lens groups, respectively. The face gasket is used to be detachably assembled to the body and used to contact a user's face. The face gasket has two adjustment knobs. Each of the adjustment knobs is used to couple a corresponding one of the adjustment rings and rotate a corresponding one of the zoom lens groups for zooming, and a part of the each of the adjustment knobs is exposed to the outside for the user to manipulate.

Based on the above, in the head-mounted display device of this application, the face gasket has adjustment knobs so that the user may perform zooming by the adjustment knobs.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic diagram of an assembled state of a head-mounted display device of an embodiment of the invention.

FIG. 2 is a schematic diagram of a disassembled state of the head-mounted display device in FIG. 1.

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FIG. 3 is a schematic diagram of an adjustment ring and an adjustment knob of the head-mounted display device in FIG. 1.

DESCRIPTION OF THE EMBODIMENTS

FIG. 1 is a schematic diagram of an assembled state of a head-mounted display device of an embodiment of the invention. FIG. 2 is a schematic diagram of a disassembled state of the head-mounted display device in FIG. 1. Referring to FIG. 1 and FIG. 2, a head-mounted display device 100 of this embodiment includes a body 110, two adjustment rings 120, and a face gasket 130. The body 110 has two zoom lens groups 112 corresponding to both eyes. The two adjustment rings 120 are used to be detachably assembled to a corresponding one of the zoom lens groups 112, respectively. The face gasket 130 is used to be detachably assembled to the body 110 and used to contact a user's face. The face gasket 130 has two adjustment knobs 132. Each of the adjustment knobs 132 is used to couple a corresponding one of the adjustment rings 120 and a corresponding one of the zoom lens groups 112 for zooming, and a part of the each of the adjustment knobs 132 is exposed to the outside for the user to manipulate.

In this embodiment, the face gasket 130 is detachably assembled to the host 110. As a result, when the user uses the head-mounted display device 100 in a static usage situation, the face gasket 130 may be removed to reduce weight in favor of another lightweight face gasket. On the other hand, when the user uses the head-mounted display device 100 in a situation where there may be intense movements, or when using the head-mounted display device 100 in an immersive situation that requires isolation from the outside world, the face gasket 130 may be assembled to the body 110, which not only improves the stability when wearing the head-mounted display device 100, but also reduces interference and provides a better immersive experience by blocking out external light.

In addition, in the head-mounted display device 100 of this embodiment, the two zoom lens groups 112 may zoom independently. When the user has a myopia problem, the zoom lens groups 112 may be zoomed to get a clear image. Moreover, even if the user's eyes have different degrees of myopia, the two zoom lens groups 112 may be individually zoomed to different degrees, so that both eyes can get a clear image. When the face gasket 130 is not assembled to the host 110, the user may directly touch and rotate the zoom lens groups 112 to perform zooming. When the face gasket 130 is assembled to the host 110, the user may still touch the part of the adjustment knobs 132 exposed to the outside for the user to manipulate, and the manipulated adjustment knobs 132 may drive the adjustment rings 120 to rotate the corresponding one of the zoom lens groups 112 for zooming. Therefore, the user may conveniently complete the zooming of the zoom lens groups 112 with or without assembling the face gasket 130 to the host 110.

In this embodiment, the host 110 is a lightweight head-mounted display similar to glasses, which may be applied, for example, in the field of virtual reality system, augmented reality system, or mixed reality system. The host 110 may include components such as an optical system and a protective casing, and may be provided with a display or be suitable for placing a display. The display may be a built-in display or an external portable display (e.g., a smartphone, etc.), but the application is not limited thereto. The optical system includes optical elements used to change the light path of the display, such as lenses, light guides, or prisms.

In this embodiment, the head-mounted display device **100** further includes a headband module **140** having two opposite connecting ends **142** used to be detachably assembled to two legs of glasses **114** of the body **110**. For example, the connecting ends **142** and the legs of glasses **114** may be connected by hook and loop fasteners or other methods. When the user uses the head-mounted display device **100** in a situation where there may be intense movements, in addition to assembling the face gasket **130** to the body **110**, the headband module **140** may also be assembled to the body **110**. In this way, the face gasket **130** and the headband module **140** may touch the front and back of the user's head respectively, providing wearing stability and comfort in the way of front and back clamping, close to the feeling of an all-in-one machine. Even if the user has a more intense movement, the head-mounted display device may still be fixed steadily on the user's head, without worrying about dropping or even damaging it.

In this embodiment, a length of the headband module **140** is adjustable. By adjusting the length of the headband module **140**, users with different head sizes may comfortably and securely wear the head-mounted display device **100** on their heads. In this embodiment, the headband module **140** further has a built-in battery **144** electrically connected to the host **110**. The host **110** itself may have a battery, but such a design also increases the weight of the host **110**. Alternatively, the host **110** could be powered by an external power source using a cable, such as a computer or a mobile power source. However, the cable may affect the user's operating experience. With the headband module **140** equipped with the battery **144**, the head-mounted display device **100** does not need to be connected to an external power source, which may avoid limiting the user's movements when the user is using the head-mounted display device **100**. Moreover, such a structure also allows the head-mounted display device **100** to be more balanced in weight on the user's head, providing better operational stability.

In this embodiment, the headband module **140** further has a communication module **146** used to receive, for example, fifth generation mobile communication signals or other wireless communication signals for communication between the host **110** and the outside world. In this embodiment, the headband module **140** may further have a head support **148** used to contact the back of the user's head to provide more stable support.

FIG. 3 is a schematic diagram of an adjustment ring and an adjustment knob of the head-mounted display device in FIG. 1. Other components of the head-mounted display device are omitted in FIG. 3 to clearly understand the coupling relationship between the adjustment rings and the adjustment knobs. Referring to FIG. 1 and FIG. 3, in this embodiment, each of the adjustment knobs **132** is a gear set. For example, each of the adjustment knobs **132** includes a

gear **132A** and a gear **132B**. A part of the gear **132A** is exposed to the outside for the user to manipulate. The gear **132A** also engages the gear **132B**, and gear **132B** engages a rack **122** on adjustment ring **120**. When the user rotates the gear **132A**, the gear **132B** is rotated by the gear **132A** and further pivots the rack **122** to rotate the adjusting ring **120**, and then the adjusting ring **120** rotates the zoom lens group **112** to perform zooming.

To sum up, in the head-mounted display device of this application, even if the face gasket is assembled to the body, the user may still perform zooming by the adjustment knobs of the face gasket, and there is no need to repeatedly remove the head-mounted display device in order to touch the zoom lens groups for zooming and then put the head-mounted display device on again to confirm whether the zoom status is ideal, which may improve the convenience of use. In addition, when combined with the detachable face gasket and headband module, the head-mounted display device may be firmly fixed on the user's head and not easily dropped. When the head-mounted display device is used in a relatively stable usage situation, the face gasket and headband module may also be removed to reduce weight, and other lightweight face gaskets may be used.

What is claimed is:

1. A head-mounted display device comprising:

a body having two zoom lens groups corresponding to both eyes;

two adjustment rings used to be detachably assembled to a corresponding one of the zoom lens groups, respectively; and

a face gasket used to be detachably assembled to the body and used to contact a user's face, wherein the face gasket has two adjustment knobs, each of the adjustment knobs is used to couple a corresponding one of the adjustment rings and rotate a corresponding one of the zoom lens groups for zooming, and a part of the each of the adjustment knobs is exposed to the outside for the user to manipulate.

2. The head-mounted display device according to claim 1, wherein the each of the adjustment knobs is a gear set.

3. The head-mounted display device according to claim 1 further comprising a headband module having two opposite connecting ends used to be detachably assembled to two legs of glasses of the body.

4. The head-mounted display device according to claim 3, wherein the headband module further has a built-in battery.

5. The head-mounted display device according to claim 3, wherein the headband module further has a communication module.

6. The head-mounted display device according to claim 3, wherein the headband module further has a head support used to contact a back of the user's head.

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